

Novel probiotic candidates for humans isolated from raw fruits and vegetables.



This study was aimed at determining the probiotic potential of a large number of autochthonous lactic acid bacteria isolated from fruit and vegetables. Survival under simulated gastric and intestinal conditions showed that 35% of the strains, mainly belonging to the species *Lactobacillus plantarum* maintained high cell densities. Selected strains did not affect the immune-mediation by Caco-2 cells. All strains stimulated all 27 immune-mediators by peripheral blood mononuclear cells (PBMC). A significant ($P<0.05$; $P<0.01$) increase of the major part of cytokines and growth factors was found. A few chemokines were stimulated. Immune-mediators with pro-inflammatory activity (IL-17, EOTAXIN and $IFN\gamma$) were significantly ($P<0.01$) stimulated by all strains, followed by IL-1b>IP-10>IL-6>MIP1 α . Stimulation of IL-12, IL-2 and IL-7 was strain dependent. Only a few strains increased the synthesis of cytokines with anti-inflammatory activity. Six *L. plantarum* strains were further selected. Four were defined as the strongly adhesive strains (more than 40 bacteria adhering to one Caco-2 cell), and 2 as the adhesive strains (5-40 bacteria adhering to one Caco-2 cell). Five strains grew and acidified chemically defined medium with fructo-oligosaccharides (FOS) as the only carbon source. End-products of FOS fermentation were found. All strains inhibited enterohemorrhagic *Escherichia coli* K12 and *Bacillus megaterium* F6 isolated from human sources. The results of this study showed that some autochthonous lactic acid bacteria from raw fruit and vegetables have functional features to be considered as novel probiotic candidates. Copyright © 2012 Elsevier Ltd. All rights reserved.